

[Feedback Control Systems]

[(Open Ended Lab) Report]



[Design of Light Intensity Control Using PID Controller]

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Introduction

This project focuses on designing a **hardware-based PID controller** to regulate the brightness of a filament DC bulb according to a user-defined reference level. The system uses a **closed-loop feedback mechanism**, where a Light Dependent Resistor (LDR) senses the actual light intensity and converts it into a voltage signal. This signal is compared with the reference input to generate an error, which is then processed by the PID controller.

The key concepts used in this project include **feedback control systems, analog signal processing, and PID control (Proportional, Integral, Derivative)**. The proportional part provides immediate response, the integral part eliminates steady-state error, and the derivative part improves system stability and response speed.

The **significance** of this project lies in demonstrating how **real-time control systems can be implemented using analog hardware**, without microcontrollers. It also helps in understanding practical challenges like sensor nonlinearity, system stability, and tuning of controller parameters. A slightly advanced idea used here is the **implementation of PID control using operational amplifiers (LM741)** instead of digital methods. This provides deeper insight into how control systems work at the hardware level and is useful in industrial environments where fast and continuous control is required.

Problem Analysis

The problem requires designing a system that can **control the light intensity of a 5V filament DC bulb** according to a user-defined reference value. The system must continuously measure the actual light intensity using an LDR sensor and adjust the bulb brightness so that it matches the desired level. The main challenge is to convert light intensity into a measurable voltage signal and then regulate the bulb in real time using only hardware components.

To solve this problem, a **closed-loop control system** will be implemented. The LDR will be used in a voltage divider circuit to convert light intensity into a voltage signal. This signal will be compared with the reference voltage to generate an error. A **hardware-based PID controller** (using op-amps, resistors, and capacitors) will process this error and produce a control signal. Finally, a **BJT transistor** will be used to drive the bulb by adjusting the current according to the PID output.

This approach ensures that the system automatically corrects any difference between desired and actual light intensity, resulting in stable and accurate control.

Design Requirements

The objective of this project is to control the intensity of a filament-based DC bulb using a hardware PID controller under given constraints.

Input:

- Reference voltage (user-defined desired light intensity)
- Feedback signal from LDR (represents actual light intensity in voltage form)

Output:

- Controlled voltage/current applied to the bulb to adjust its brightness

Constraints:

- Bulb operates at **5V and draws up to 500mA**, so the driver circuit must handle this safely
- Use only available components: **LM741 op-amps, BJTs, resistors, and capacitors**
- LDR provides a **nonlinear response**, so proper signal conditioning is required
- System should be stable with **minimal oscillations and fast response**
- Output must not exceed the bulb's voltage rating

The system must continuously compare input and feedback, process the error through the PID controller, and adjust the output to maintain the desired light intensity.

Feasibility Analysis

The feasibility of this project is evaluated in terms of **time management** and **cost management** to ensure that the desired objectives can be achieved within given constraints.

Time Management:

The project is feasible within the allocated time as it involves well-understood components such as op-amps, BJTs, LDR, and passive elements. The design can be divided into stages including circuit design, simulation, hardware implementation, and testing. Since the PID controller is implemented using standard analog circuits, it does not require complex programming, which reduces development time. Proper planning and step-by-step testing ensure timely completion.

Cost Management:

The project is cost effective as all components used are low cost and readily available in laboratories or local markets. Components like LM741 op-amps, resistors, capacitors, LDR, and BJTs are inexpensive. No expensive microcontrollers or specialized sensors are required, which keeps the overall cost minimal.

Overall Feasibility:

Considering both time and cost factors, the project is highly feasible. It can be successfully implemented using available resources while meeting the required performance and design constraints.

Possible Solutions

There are multiple approaches to control the light intensity of a DC bulb:

1. Open-Loop Control:

In this method, the bulb is driven directly based on input without feedback.

- *Advantage:* Simple design
- *Disadvantage:* No correction for disturbances, poor accuracy

2. Microcontroller-Based PID Control:

Uses a microcontroller to implement PID digitally.

- *Advantage:* High accuracy, easy tuning, flexible
- *Disadvantage:* Higher cost, requires programming, not purely hardware-based

3. Analog PID Control (Op-Amp Based):

Implements PID using op-amps, resistors, and capacitors.

- *Advantage:* Real-time response, no programming required, fulfills hardware constraint
- *Disadvantage:* Requires careful tuning, less flexible than digital

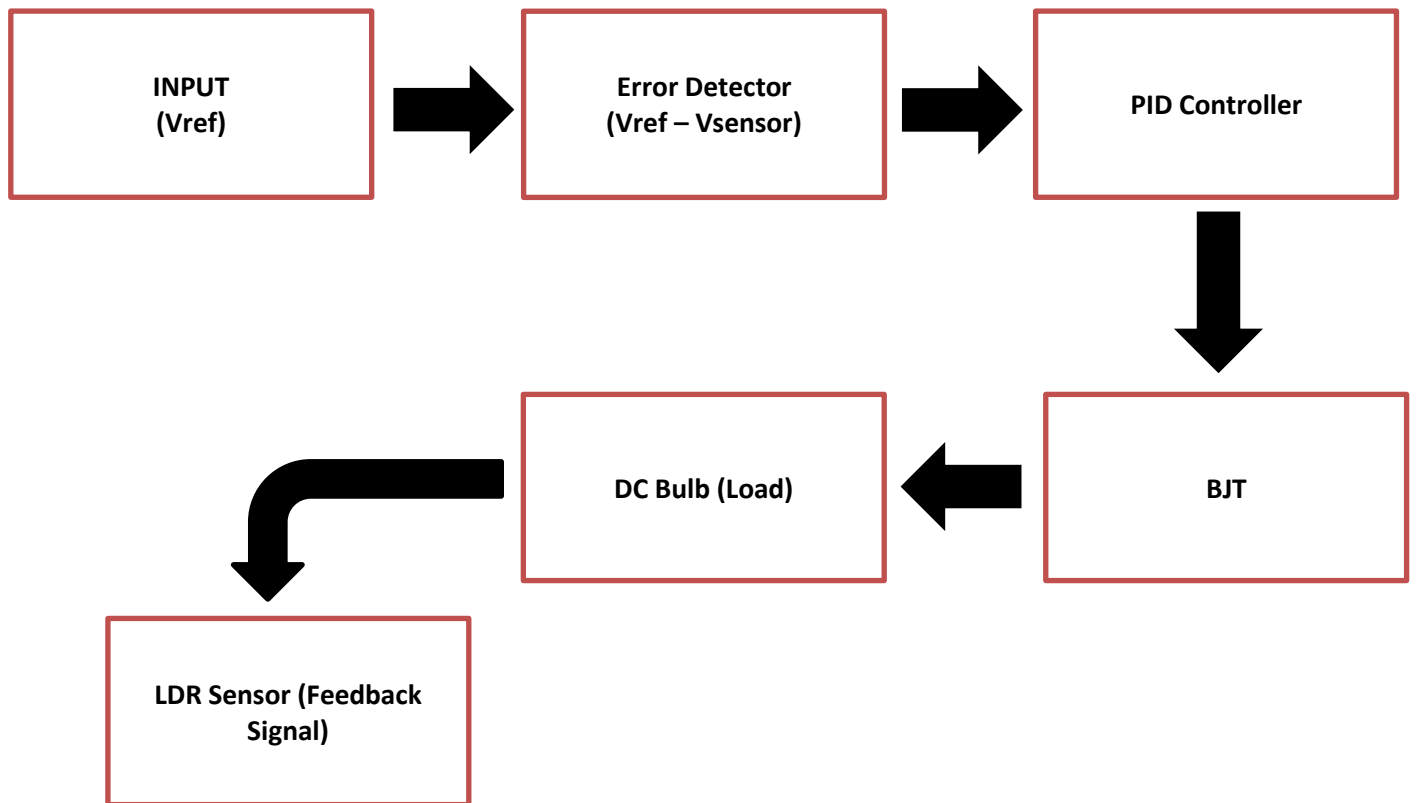
Selected Solution and Justification

The **analog PID controller using op-amps** is selected as the best solution.

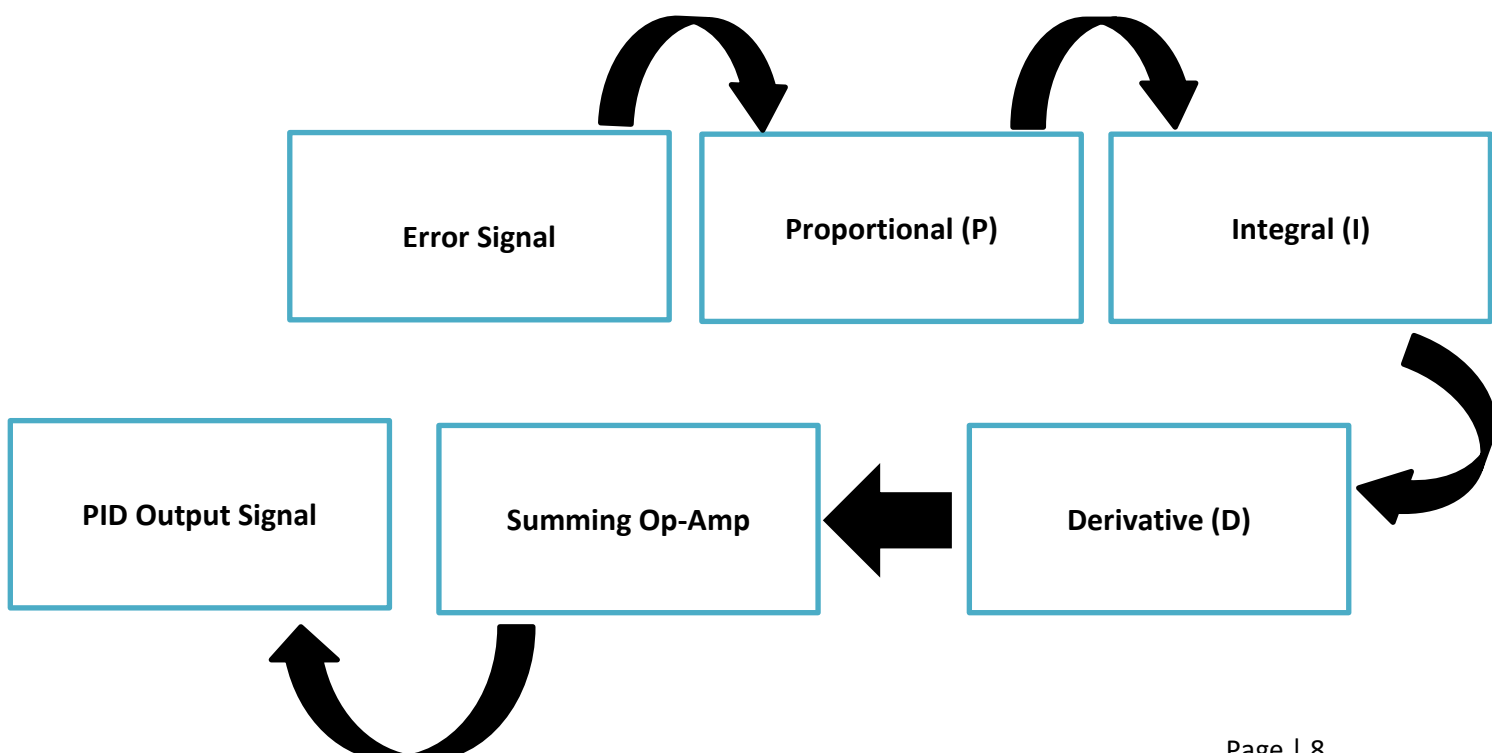
Reason for selection:

- Provides closed-loop control with good stability and accuracy
- Uses low-cost and readily available components

Preliminary Design



PID Controller Block Diagram



Design Description

1. Reference Input

This block provides the **desired light intensity level** set by the user. It is given as a reference voltage (V_{ref}). This voltage acts as the target value that the system tries to maintain by controlling the bulb brightness.

2. LDR Sensor

The Light Dependent Resistor (LDR) is used to sense the actual light intensity. Its resistance decreases when light increases and increases when light decreases. It is connected in a **voltage divider configuration** to convert resistance changes into a measurable voltage (V_{sensor}). This voltage represents the real-time light intensity of the bulb.

3. Error Detection

This block compares the reference input (V_{ref}) with the sensor output (V_{sensor}). The difference between them is called the **error signal**:

- If light is low $\rightarrow$ error is high $\rightarrow$ bulb intensity increases
- If light is high $\rightarrow$ error is low $\rightarrow$ bulb intensity decreases

This error signal is the main input to the controller.

4. PID Controller (Op-Amp Based)

This is the main control unit implemented using **LM741 operational amplifiers**.

It consists of three parts:

- **Proportional (P)**: Produces output proportional to error, improving response speed
- **Integral (I)**: Removes steady-state error by accumulating past error
- **Derivative (D)**: Predicts future error changes and improves stability

These three outputs are combined using a **summing amplifier**, producing a final control signal.

Design Description

5. Power Driver Block (BJT Stage)

The PID output is not strong enough to drive the bulb directly, so a **BJT transistor (e.g., TIP122 or equivalent)** is used as a power amplifier.

- It amplifies current from the PID output
- Controls the power delivered to the bulb
- Ensures safe operation for high current load (500mA bulb)

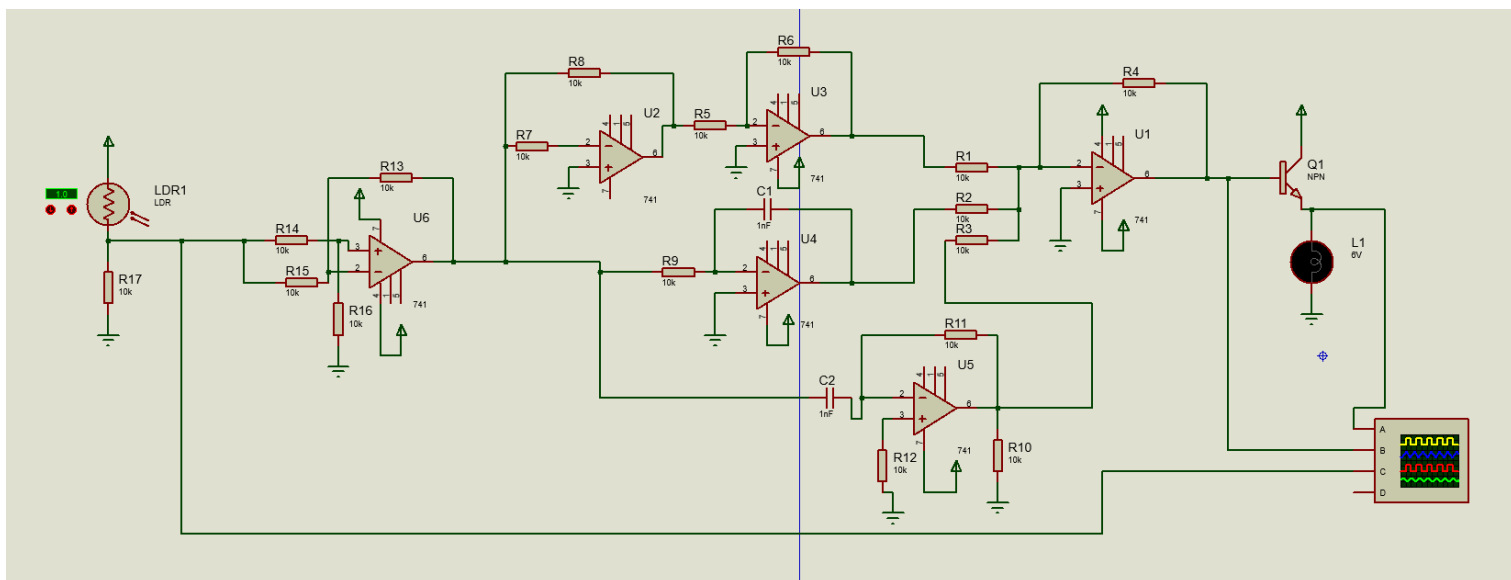
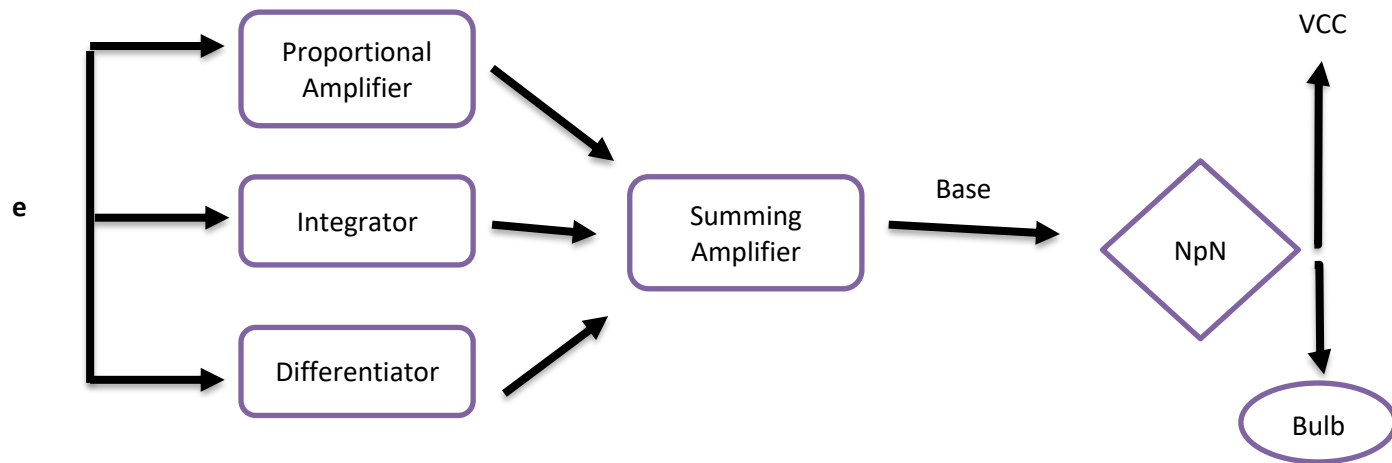
6. DC Bulb (Output Load)

The filament-based DC bulb acts as the **final controlled system output**. Its brightness changes depending on the current supplied by the BJT driver. This is the actual physical parameter being controlled.

7. Feedback Loop

The light produced by the bulb is again sensed by the LDR. This creates a **closed-loop feedback system**, continuously adjusting the output to match the reference value. This loop is essential for maintaining stable and accurate control.

Software Simulation (Optional)



Experimental Results

Observations / Results Obtained

After implementing the hardware-based PID control system, the following results were observed during testing:

- The system successfully regulated the **brightness of the DC bulb** according to the reference voltage set by the user.
- When the reference light level was increased, the bulb brightness increased smoothly, and when it was decreased, the brightness reduced accordingly.
- The **LDR feedback effectively tracked changes in light intensity**, producing corresponding voltage variations in real time.
- The PID controller improved system behavior by:
 - Reducing steady-state error (due to integral action)
 - Improving response speed (due to proportional action)
 - Minimizing oscillations and overshoot (due to derivative action)

Oscilloscope Observations

- The output waveform showed a **stable settling behavior** after initial transient response.
- Error signal decreased gradually and reached near-zero in steady state.
- The bulb driving voltage became stable after a short adjustment period.

Overall Result

The system achieved **effective closed-loop control of light intensity**, maintaining the bulb brightness close to the desired reference value with good stability and reduced fluctuations.

Performance Analysis

The implemented hardware-based PID controller shows effective performance in regulating the light intensity of the DC bulb. The system operates as a closed-loop feedback control system, continuously adjusting the output based on the difference between the reference and measured light intensity.

The **proportional term** provides an immediate response to changes in error, ensuring that the system reacts quickly when the light level deviates from the desired value. However, using proportional control alone would result in a steady-state error.

The **integral term** successfully eliminates this steady-state error by accumulating past errors and forcing the system output to reach the exact reference level over time. This improves overall accuracy but may slightly slow down the response if not properly tuned.

The **derivative term** improves system stability by predicting the rate of change of error. It helps reduce overshoot and minimizes oscillations in the output response, making the system smoother and more stable.

Overall, the system shows:

- Good **tracking ability** of reference light intensity
- Reduced **steady-state error**
- Improved **stability with minimal oscillations**
- Reasonable **response time under changing conditions**

However, some limitations were also observed:

- LDR non-linearity affects measurement accuracy
- Manual tuning of PID parameters is required for optimal performance
- LM741 introduces slight delay and noise sensitivity in practical circuits

Conclusion

The designed hardware-based PID controller successfully achieved the objective of regulating the intensity of a DC filament bulb using LDR based feedback. The system operated effectively as a closed loop control system, where the light output was continuously adjusted according to the user defined reference value.

From the results and performance analysis, it is concluded that the combination of proportional, integral, and derivative actions significantly improves system behavior. The **proportional term** ensures fast response, the **integral term** removes steady-state error, and the **derivative term** enhances stability by reducing oscillations. Together, they provide smooth and accurate control of light intensity.

The implementation using LM741 op amps and BJTs demonstrated that complex control systems like PID can be successfully realized using simple analog components without microcontrollers. Although some limitations such as sensor non-linearity and manual tuning were observed, the overall system performance remained stable and effective.

In conclusion, the project successfully demonstrates a practical application of analog control theory and confirms that hardware based PID control is a feasible and efficient approach for real-time light intensity regulation under given constraints.

References

- [1] Nise, N. S. — *Control Systems Engineering*, Wiley.
- [2] Sedra, A. S., & Smith, K. C. — *Microelectronic Circuits*, Oxford University Press.
- [3] Boylestad, R. L., & Nashelsky, L. — *Electronic Devices and Circuit Theory*, Pearson.
- [4] LM741 Operational Amplifier Datasheet, Texas Instruments.
- [5] Basic Analog Electronics Laboratory Manual — Control Systems and Op-Amp Applications.
- [6] Light Dependent Resistor (LDR) Technical Datasheet — Standard Sensor Reference.

Range of Complex Engineering Activity

EA1: Range of resources EA2: Level of interaction EA3: Innovation EA4: Consequences for society and environment EA5: Familiarity	• EA1: Range of resources -- The design involves the use of diverse resources, such as, money, equipment, information, and technology. • EA2: Level of Interaction – The design requires resolution of various problems arising from interactions between wide-ranging or conflicting technical or other issues. • EA3: Innovation – The design addresses sustainability and reduces cost requiring creative use of engineering principles and research-based knowledge in novel ways. • EA5: Familiarity – The design activity emphasizes the integration of existing knowledge and tools (or familiar solutions) with new or unfamiliar challenges			
	Rubrics		LLOs	Marks
	Explains the design process including engagement of resources clearly	EA1	LLO3	
	Demonstrate the final design clearly with all supporting information	EA1, EA5	LLO4	
	Demonstrate how innovation has been used in design to resolve conflicting requirements including the impact on society and environment	EA2, EA3	LLO4	
	Prepares a report which explain the design process including engagement of resources clearly, and is free from grammatical errors.	EA1	LLO3	

